

Computational power required to fine tune a LLM/SLM

Posted by u/Mother-Proof3933 • • 0 points • 0 comments

Hey all,

I have access to 8 A100 -SXM4-40 GB Nvidia GPUs, and I'm working on a project that requires constant calls to a Small Language model (phi 3.5 mini instruct, 3.82B for example).

I'm looking into fine tuning it for the specific task, but I'm unaware of the computational power (and data) required.

I did check google, and I would still appreciate any assistance in here.

Comments

u/BenniB99 • 1 points • 2025-04-02 11:00 UTC

You could train 70B+ models with 8 40GB GPUs.

For instance with [unsloth](<https://docs.unsloth.ai/get-started/beginner-start-here/unsloth-requirements>) you would only need around 8-10GB of VRAM (depending on hyperparameters and context size) to train a 3-4B model with LoRA, note that they do not support multiple gpus (yet).

For training bigger models on multiple gpus you will probably want to use something like transformers and trl from huggingface right now.

In terms of data, I have had good results with LoRA and a couple hundred samples.

I think a good starting point would be to just use a PEFT method and see how far it gets you.

u/microcandella • 1 points • 2025-04-04 20:48 UTC

What time ranges would it take to do this?

u/BenniB99 • 2 points • 2025-04-04 21:38 UTC

You mean the amount of time finetuning takes?

Not that long for only a couple hundred rows, e.g. training for 3 epochs on a RTX 3090 for a dataset with 300 samples and a batch size of 1 usually takes around an hour for a 8B model.

u/microcandella • 1 points • 2025-04-04 22:17 UTC

Thanks! That's much faster than I thought.